

Carbonized rice husks as absorber materials for solar-driven steam generation devices

Thi Hien Nguyen^a, Nam Anh Tran^{a,b}, Thi Bich Ngoc Nguyen^a, Thi My Duyen Bui^a,
Phuong Anh Lam^c, Ho-Suk Choi^d, Dieu Linh Do^e, Thi Hanh Nguyen^b, Viet Phuong Hoang^a,
Hai Linh Nguyen^a, Thuy Duong Pham^a, Van-Duong Dao^{a,*}

^a Faculty of Biotechnology, Chemistry and Environmental Engineering, PHENIKAA School of Engineering, PHENIKAA University, Hanoi, 10000, Viet Nam

^b Faculty of Environmental Sciences, University of Science, Vietnam National University, 334 Nguyen Trai, Hanoi, 10000, Viet Nam

^c Department of Chemistry, Case Western Reserve University, 10900 Euclid Avenue, Cleveland, OH, 44106, United States

^d Department of Chemical Engineering and Applied Chemistry, Chungnam National University, Daejeon, 34134, South Korea

^e HUS – Hightschool for the Gifted Students, Thanh Xuan, Hanoi, 10000, Viet Nam

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ABSTRACT

Using biomass-based carbon photothermal materials in solar-driven steam generation (SSG) has recently become an outstanding topic. The synthesis of carbon materials from agricultural residues and their applications in photothermal energy conversion and clean water production have emerged as highly promising and sustainable research directions. In this study, we focus on solar steam generation (SSG), which integrates both photothermal energy conversion and clean water production into a unified application. Rice husk (RH), an agricultural waste product rich in carbon content, was utilized to synthesize a low-cost carbon-based photothermal material for SSG devices. Characterization techniques, namely XRD, FTIR, EDS, SEM, and UV-Vis DRS, were employed to investigate the properties of the obtained material. With excellent absorption capability across the full range of solar radiation, and high hydrophilicity, rice husk-based carbon (RHC) illustrates the great potential for producing steam from solar energy. Under standard solar conditions, the RHC-based SSG system achieved a steam generation rate of $2.09 \text{ kg m}^{-2} \text{ h}^{-1}$ at an ambient temperature of 25°C and humidity of 60 % with a thickness of $132 \mu\text{m}$. Additionally, RHC shows a removal rate of ions above 85 % during purification, which can provide a great way for real-world applications to produce clean water from seawater and freshwater.

1. Introduction

Using solar irradiation is one of the smartest and most effective approaches to solving energy and environmental problems. So far, many studies have been published on harnessing solar energy for electricity and thermal production, hydrogen energy, or carbon-based fuels [1–10]. These methods are also used in processes to treat organic pollutants in the aquatic environment through direct and indirect photolysis [11–13]. In addition, one of the newest and most promising photo-thermal applications that should be mentioned is solar-driven steam generation (SSG). The related applications include desalination, wastewater treatment and purification, disinfection, and clean-water production

[14–20]. The performance of SSG is influenced by several factors such as thermal management, water transportation, and water evaporation. To date, two designs of SSG systems have been developed, namely the volumetric system and the floating system. Although volumetric systems are easy to fabricate because they only require dispersing the materials into water, they also have several drawbacks, such as thermal energy loss during bulk water heating. Additionally, bulk water and other components can also prevent light from reaching the materials. Moreover, to disperse hydrophobic materials, a modification step may be required. In contrast, floating systems can achieve a high evaporation rate, but they have a complex structure. Furthermore, the material's adhesive ability to form the absorber layer remains a challenge.

* Corresponding author.

E-mail address: duong.daovan@phenikaa-uni.edu.vn (V.-D. Dao).

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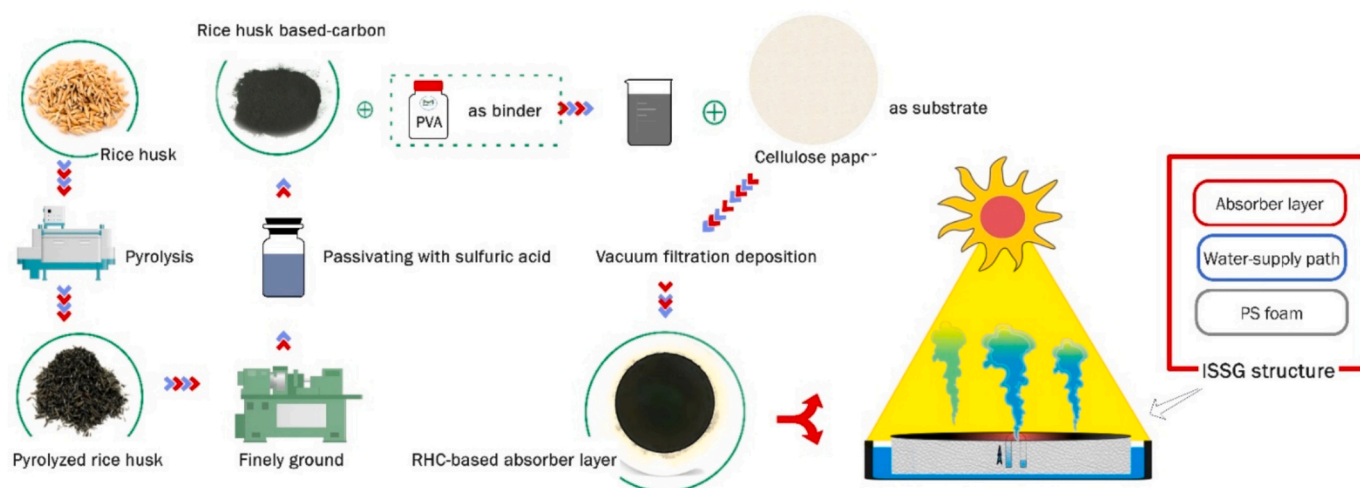


Fig. 1. Experimental scheme for fabricating solar absorption films from RHC.

In addition to the SSG structure, the steam generation rate is directly related to the photothermal material. Two groups of materials with broadband solar absorption, namely metallic-based and carbon-based nanoparticles, are usually employed in SSG [21,22]. Although highly efficient SSG systems can be achieved, the use of metals results in high costs and limitations at a large scale. In contrast, carbon-based materials are cheap and easy to fabricate.

Minimizing investment costs and ensuring high work efficiency are always the goals of both researchers and R&D companies [10,23]. This problem can be solved by nature-inspired design for effective water evaporation driven by solar energy [24]. Carbonization of natural materials such as daikon [25], lotus [26], sunflower heads [27], mushrooms [28], lotus seedpods [29], and rice husk [19,30,31] has been applied. Among them, rice husk, a common agricultural waste, has also been reported to be an effective solar absorber [30–33]. Wei Fang et al. used foaming and freeze-drying, the resulting 3D carbonized rice husk foam SSG system provided a steam generation rate of $1.0315 \text{ kg m}^{-2} \text{ h}^{-1}$. It should be noted that their research did not assess the applicability of rice husk-based materials in volumetric solar-driven steam generation. On the other hand, the fabrication process took a long time and included many complex steps, such as foaming and freeze-drying [30].

Vietnam is an agricultural country where rice serves as both a staple food and a national symbol. A significant amount of rice husk waste is generated annually, which has motivated scientists to explore ways to process this agricultural byproduct. Therefore, we have chosen rice husk to synthesize the rice husk-based carbon (RHC) for application in solar steam generation (SSG) systems. Notably, two RHC-based SSG systems, namely volumetric and floating, were evaluated and compared. Specifically, in the floating system, the absorption layer was fabricated by coating RHC onto cellulose. The results indicated that the floating structure exhibited significantly higher performance, achieving a UV–Vis light absorption rate of over 95 %. The solar-driven steam generation device, designed with an absorber layer on top of polystyrene foam and a cellulose bridge as a water supply path, achieved a high evaporation rate of $2.09 \text{ kg m}^{-2} \text{ h}^{-1}$ at a power density of 1 sun (1 kW m^{-2}).

2. Materials and methods

2.1. Preparation of RHC-based interfacial solar steam generator (RHC-based ISSG)

Rice husk (RH) was harvested from a rice farm in Hanoi, Vietnam (Khang Dan Rice), and cleaned to remove dirt and dust particles using deionized water. RH was first dried at 100°C for several hours. Next, it was pyrolyzed at 500°C for 1 h in a tube-type electric furnace under a nitrogen atmosphere. The product was retrieved from the pyrolysis chamber once the temperature had dropped to room temperature ($25\text{--}30^\circ\text{C}$) inside the muffle furnace. Subsequently, carbon powders were ground using a high-energy ball grinding method and were modified with sulfuric acid by refluxing for 1 h at $100\text{--}120^\circ\text{C}$. The post-activated material was rinsed multiple times with deionized water until a neutral pH was achieved, and subsequently dried at 60°C . After that, an amount of RHC was added to 20 mL of PVA solution (10 g/L) and then stirred in a 70°C water bath at 60 rpm for 90 min to form a homogeneous aqueous dispersion. This solution was deposited onto the surface of a cellulose substrate (CAT No. 1442–110, GE Healthcare Life Sciences) by vacuum-assisted filtration. The RHC-coated cellulose paper was obtained after drying at atmospheric pressure for 24 h and was subsequently used as the absorber layer. Fig. 1 shows the experimental scheme for fabricating a solar absorber layer from rice husks. The interfacial solar steam generator was fabricated by placing the absorber layer on polystyrene (PS) foam, and a piece of cellulose paper was used as a water supply path.

2.2. Characterization

The surface morphology of the RHC sample was characterized using FESEM (JEOL JSM 7000F). FTIR (FTIR-4600, JASCO, Japan) and XRD (MiniFlex 600, Rigaku, Japan) were utilized to analyze the chemical structures of the materials. The optical reflectance and transmittance spectra of the RHC-coated cellulose paper were measured over the spectral range of $200\text{--}2250 \text{ nm}$ using a UV–Vis spectrophotometer (Shimadzu UV-2600). The surface temperature of the film samples was examined using a thermal imaging camera (Testo 875i IR).

2.3. Evaluation the performance of solar-driven vaporization

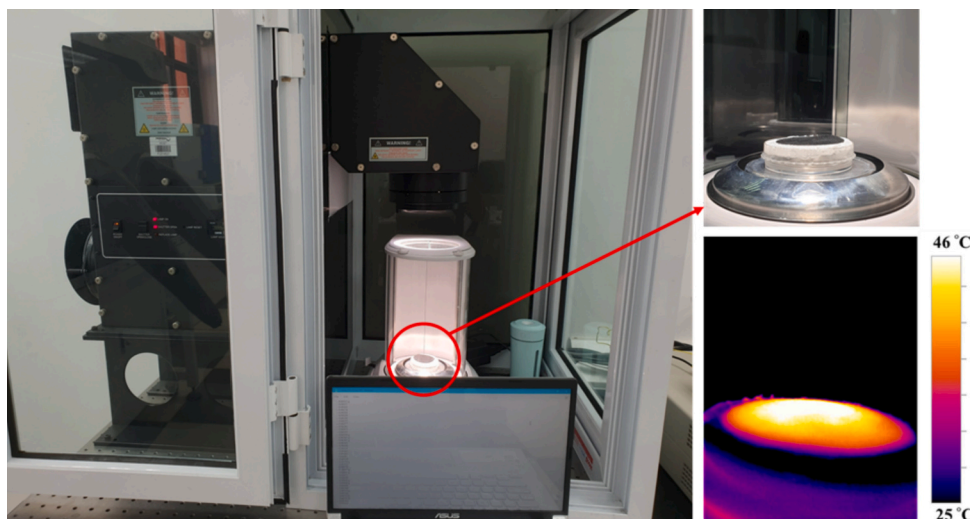


Fig. 2. Experimental design to determine the steam generation rate of RHC-based systems.

The evaporation rate of the SSG device can be calculated through weight loss of bulk water (Eq. (1)). Therefore, a high-accuracy electrical balance (ES 125SM, Precisa, Switzerland) was used for real-time monitoring of the mass changes in water. A solar simulator system (94023A Solar Sim, Newport Corporation, USA) with an AM 1.5G filter was used as the light source. The environmental conditions for all experiments were maintained at an ambient temperature of 25 °C and a humidity of 60 %. Several factors were investigated to evaluate their influence on the evaporation rate: the type of SSG system (volumetric or interfacial solar-driven steam system), the RHC coating thickness, the evaporation time, the irradiation intensity, and the number of cycles. Additionally, the concentration of ions in bulk water and condensed steam was determined by ICP-MS.

$$\nu = \frac{W_{\text{loss}} \cdot 3600 \cdot 10000}{t \cdot A \cdot 1000 \cdot 1000} \quad (1)$$

Where: W_{loss} represents the mass loss of water, A is illuminated surface area (cm^2), and t is the time of evaporation (s).

3. Results and discussion

The XRD pattern and EDS spectrum of the RHC are illustrated in Fig. 3a and b. The results of these analyses suggest the existence of amorphous carbon and monoclinic SiO_2 in the sample. In addition, a small amount of S (0.77 %) in RHC, which was introduced during the modification process with H_2SO_4 , may exist in the form of sulfonate surface groups (Fig. 3b, S-O stretch at 795 cm^{-1}). As a result, light absorption and hydrophilicity were enhanced. Additionally, the peaks spanning from 3604 to 3788 cm^{-1} correspond to the vibration of the O—H group [34]. The peak located at 2367 cm^{-1} was assigned to C=O which was attributed to carboxyl functional groups [35]. Another peak at 1612 cm^{-1} was associated with the stretching vibration of C=C. Strong vibrations of the Si-O-Si and Si—O stretching were observed at 1100 and 466 cm^{-1} , respectively. It is important to mention that the number of abundant hydrophilic groups increased significantly in the H_2SO_4 -modified carbon. These groups boosted wetting properties, which led to a fast water transfer rate of the RHC material. In particular, the presence of C=C bonds on the surface of the RHC sample increases the light-to-heat conversion efficiency of the membrane and promotes

rapid steam generation of the material [36]. The shape and size of RHC were observed by SEM (Fig. 3c). RHC exhibits a heterogeneous morphology, ranging in size from 10 to $100 \mu\text{m}$. The arrangement of

RHC on the surface of cellulose paper is shown in Fig. 4b. Initially, cellulose paper had a randomly oriented network structure of cellulose fibers. Additionally, these fibers are also stacked, creating pores and gaps of different sizes. Meanwhile, RHC particles are randomly attached to the cellulose fibers during vacuum filtration. Thus, carbon nanoparticles seem unsuitable because they can block pores and hinder the movement of water molecules. By contrast, the arrangement of micrometer-sized RHC on cellulose paper also has the advantage of creating ideal micro water pumping channels to effectively transport water to the hot regions, leading to rapid steam generation. Furthermore, the rough surface of RHC-coated cellulose paper also increases photo absorption.

Table 1

Evaporation rates of pure water, the cellulose membrane, and RHC-based SSG under dark and illuminated conditions.

Evaporation rate	Pure water	Suspending system	Cellulose membrane	Floating system
Evaporation rate under illumination ($\text{kg m}^{-2} \text{ h}^{-1}$)	0.30	0.55	0.68	1.11
Dark evaporation rate ($\text{kg m}^{-2} \text{ h}^{-1}$)	0.25	0.29	0.31	0.60

The application of carbon photothermal materials in SSG systems can be divided into two main categories: the suspending system (nanofluid/volumetric system) and a floating system (interfacial system) [37]. Therefore, we conducted experiments to study the water evaporation efficiency of RHC in both systems [38]. The evaporation rates were found to be 0.30 , 0.55 , 0.68 , and $1.11 \text{ kg m}^{-2} \text{ h}^{-1}$ for SSG devices with pure water, the suspending system, cellulose membrane (RHC-uncoated), and a floating system (RHC- $0.08 \text{ g/cellulose paper/PS foam}$), respectively, under 0.65 -sun illumination. The differences in evaporation rates can be attributed to thermal management within the structures of the evaporation system. Specifically, a floating system can minimize heat loss better than a volumetric system, as reported in recent research. Additionally, other factors that can affect the water evaporation rate include light absorption, the thickness of the RHC layers, the solar intensity, and the evaporation time [39].

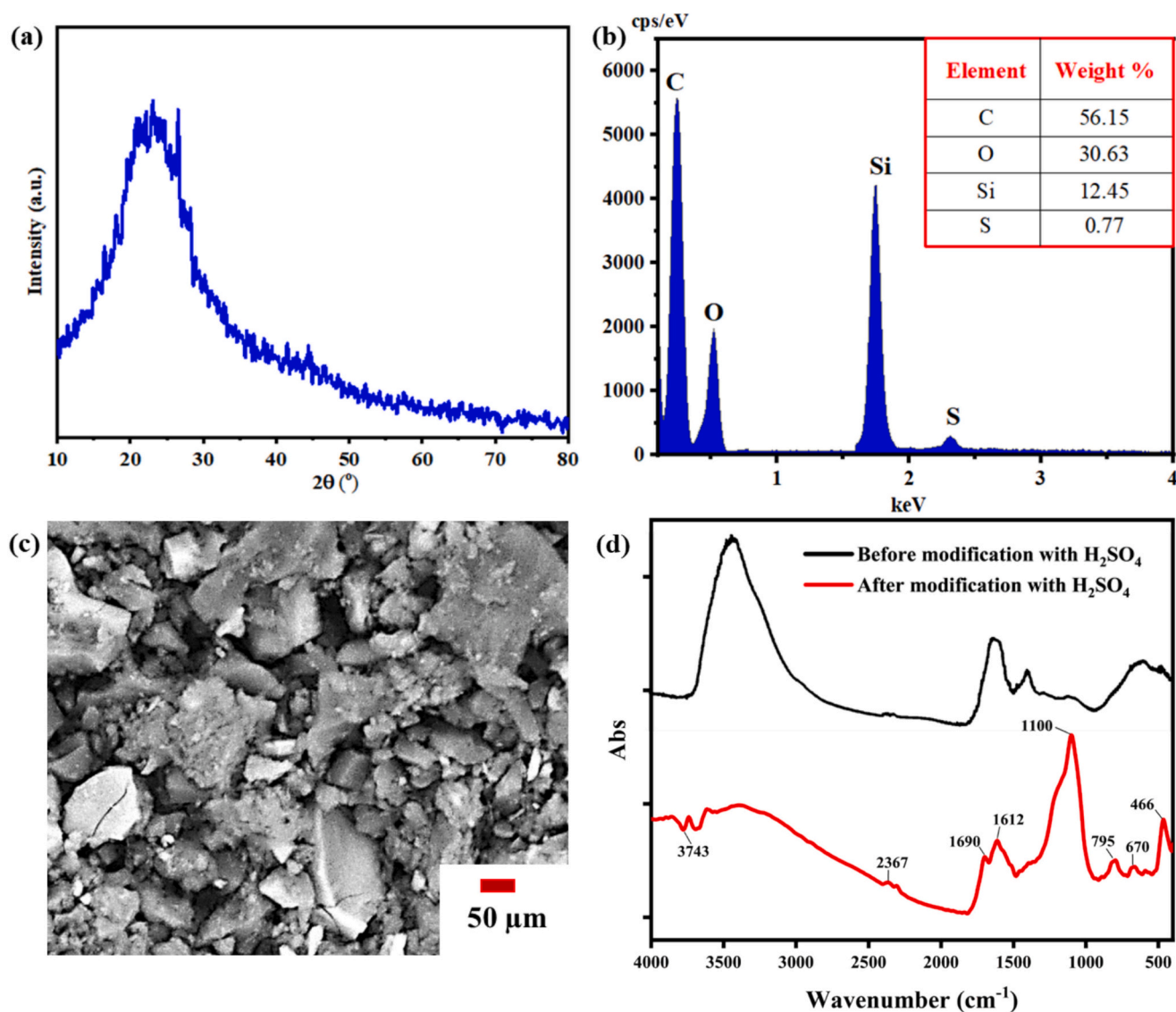


Fig. 3. Structural and morphological characterization of the RHC sample: (a) XRD pattern, (b) EDS spectrum, (c) SEM image, and (d) FTIR spectra of the carbon material obtained from pyrolysis before and after modification with H_2SO_4 .

The thickness and density of the RHCs on the exposed surface will determine the water pumping speed and the optimal water thickness on the top surface of the RHC layer, both of which significantly impact the optimization of the steam generation rate in SSG systems. To investigate this, we changed the dose of the RHC from 0.02, 0.04, 0.06, 0.08, 0.1, and 0.12 g during the filtration process. The RHC film thicknesses increased with the RHC content, as shown in Fig. 5a. As a result, we found that each additional 0.02 g of RHC increased the thickness of the RHC layer by 30 μm , confirming the reliability of the fabrication method.

An essential factor commonly used to investigate the light absorption ability of these photothermal materials is their optical properties. To this end, we recorded the optical absorption spectra using a UV-Vis-NIR spectrophotometer (Fig. 5b). Note that the light absorption of the absorber layer is not affected by changes in RHC thickness [39]. As shown in Fig. 5b, the RHC-coated paper demonstrates a broad absorption profile from ultraviolet to near-infrared wavelengths (200–2250 nm). We observed that the film with a dose of 0.08 g RHC exhibits high light absorption of over 85 % across the entire region of interest, indicating rapid steam generation and meeting the requirements for an

efficient solar absorber [37].

For investigating the solar steam conversion performance, the mass changes, evaporation rate, and surface temperature of the RHC-based absorber layer were rigorously investigated under a solar intensity of 0.65 kW m^{-2} . Fig. 6a shows the time-dependent change in water mass resulting from the steam generation process. As shown, increasing the thickness of the RHC layer also increases the variation in water mass under both irradiation conditions (dark and 0.65 sun). Fig. 6b shows the rate of water evaporation for all devices with different RHC thicknesses. From observations, the decrease in water weight of all samples under 0.65 sun is higher than that under 0 sun (dark condition). Specifically, under 0.65 sun, the steady-state evaporation rate of pure water is $0.30 \text{ kg m}^{-2} \text{ h}^{-1}$, and the water evaporation rates for RHC-based ISSG systems with RHC thickness of 0, 29, 62, 94, 132, 158, and 184 μm were 0.69, 0.88, 0.88, 0.95, 1.11, 1.01, and $0.85 \text{ kg m}^{-2} \text{ h}^{-1}$, respectively.

The surface temperature was investigated to clarify the difference in evaporation rate (Fig. 6c). The highest average surface temperature was recorded for an RHC-based ISSG system with a thickness of 184 μm (corresponding to 0.12 g RHC), at 45.5°C . The remaining absorber layers reached a stable temperature at about 42.5°C after 1500 s under

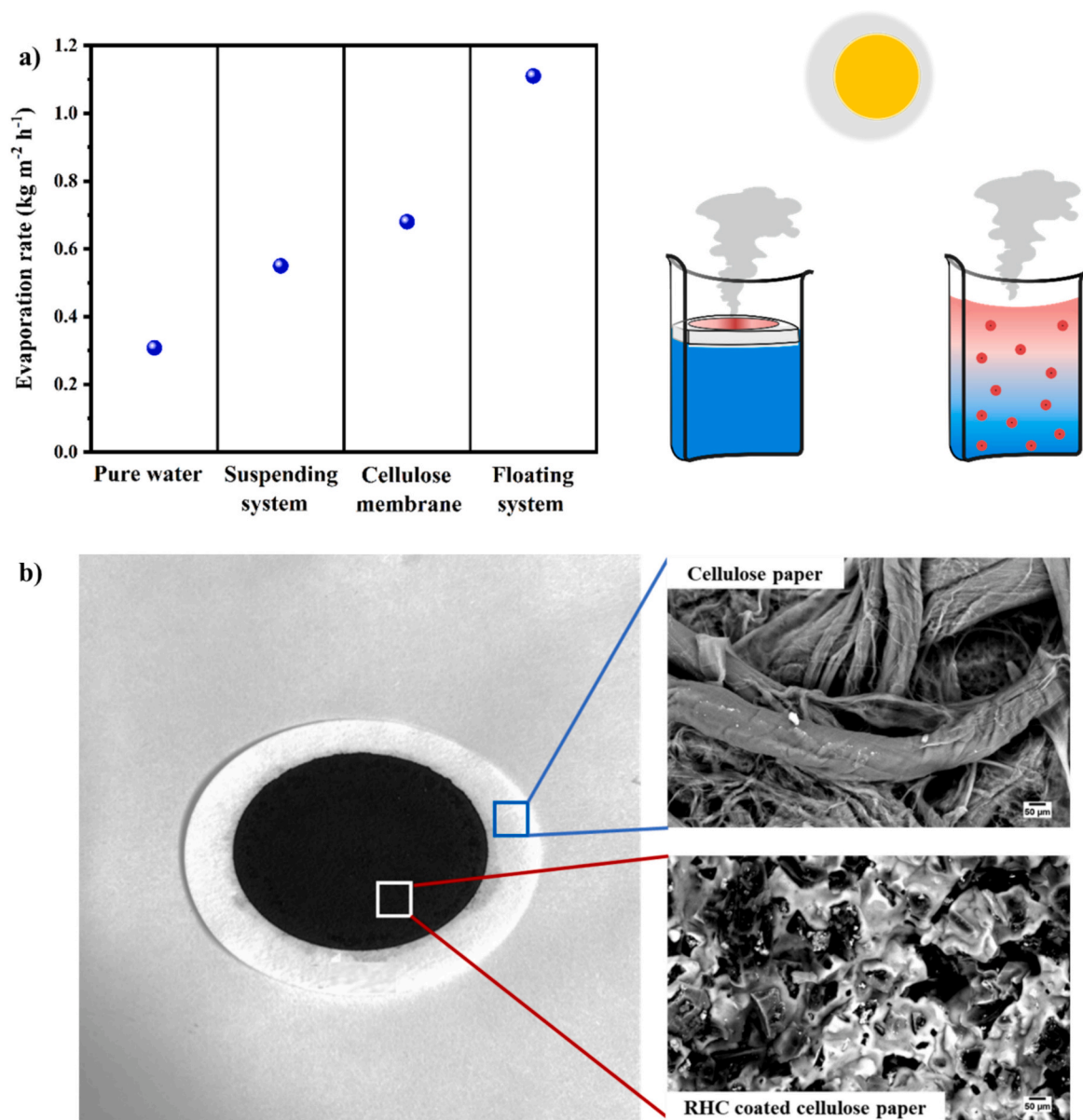


Fig. 4. (a) Evaporation rates of pure water, the cellulose membrane, and the RHC-based SSG; and (b) Surface morphology of cellulose paper and RHC-coated cellulose paper.

0.65 sun irradiation. A small volume of water on the surface of the absorber layer can cause the temperature to decrease while evaporation continues. In contrast, if water molecules cannot reach the surface of the absorber layer in time, the temperature increases and the evaporation process is interrupted due to an insufficient water supply.

The results show that the rate of water-to-steam conversion increases with light intensity (Fig. 7b). Within the limitations of the experimental equipment, the highest light intensity (1 sun) was used, and the evaporation rate reached $2.09 \text{ kg m}^{-2} \text{h}^{-1}$. Meanwhile, the water evaporation rate of the RHC-based ISSG system achieved $0.79 \text{ kg m}^{-2} \text{h}^{-1}$ under a light intensity of 0.45 sun, which is nearly three times greater than that of pure water under 0.65 sun (approximately $0.30 \text{ kg m}^{-2} \text{h}^{-1}$). These results indicate that steam generation from the RHC-based ISSG system provides a high rate of freshwater generation throughout the day. This has many implications for practical applications, especially in regions with lower solar energy availability. The evaporation rates of solar-driven steam-generation systems with different materials in recent years are summarized in Table 2.

For the practical application of photothermal materials, stability

over time is an important factor. Therefore, to assess the photothermal conversion stability of RHC and the long-term steam generation ratio of the SSG system, the steam generation recyclability of the RHC thickness of $132 \mu\text{m}$ was tested. The prolonged evaporation tests were carried out within 20 min of irradiation under 0.75 sun for each cycle. The solar steam production rates of the 1st, 2nd, 3rd, 4th, and 5th runs are 1.47, 1.46, 1.46, 1.48, and $1.47 \text{ kg m}^{-2} \text{h}^{-1}$, respectively. Over five cycles, the RHC-based ISSG system demonstrated a relatively stable evaporation rate was $1.47 \text{ kg m}^{-2} \text{h}^{-1}$, with only a slight reduction. The rate of steam generation tends to decrease with an increase in evaporation time and cycle number, which can be explained by the synthesis of light-absorbing films using PVA as a material binder on the surface of the film, the process of water transportation for a long time can affect the PVA binder, weakening the adhesion between the RHC and the cellulose paper. This can lead to loss of material or the movement of material particles in the operation process, resulting in a decrease in the rate of evaporation. Nevertheless, these results demonstrate that RHC is a photothermal material with excellent stability.

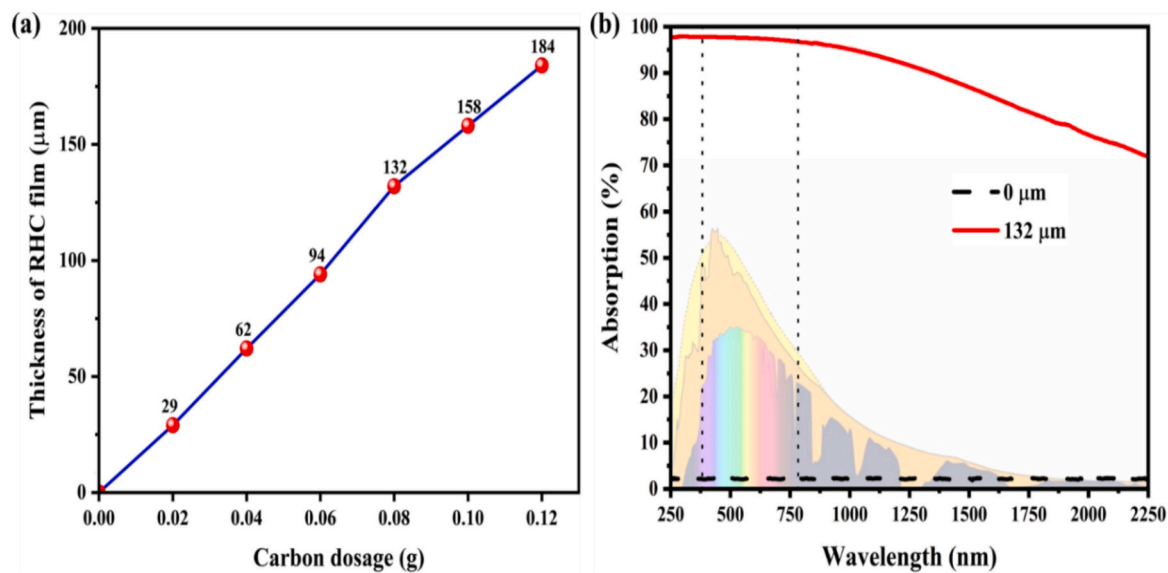


Fig. 5. (a) Thicknesses of RHC films with increasing RHC content, (b) Results of UV-Vis DRS analysis.

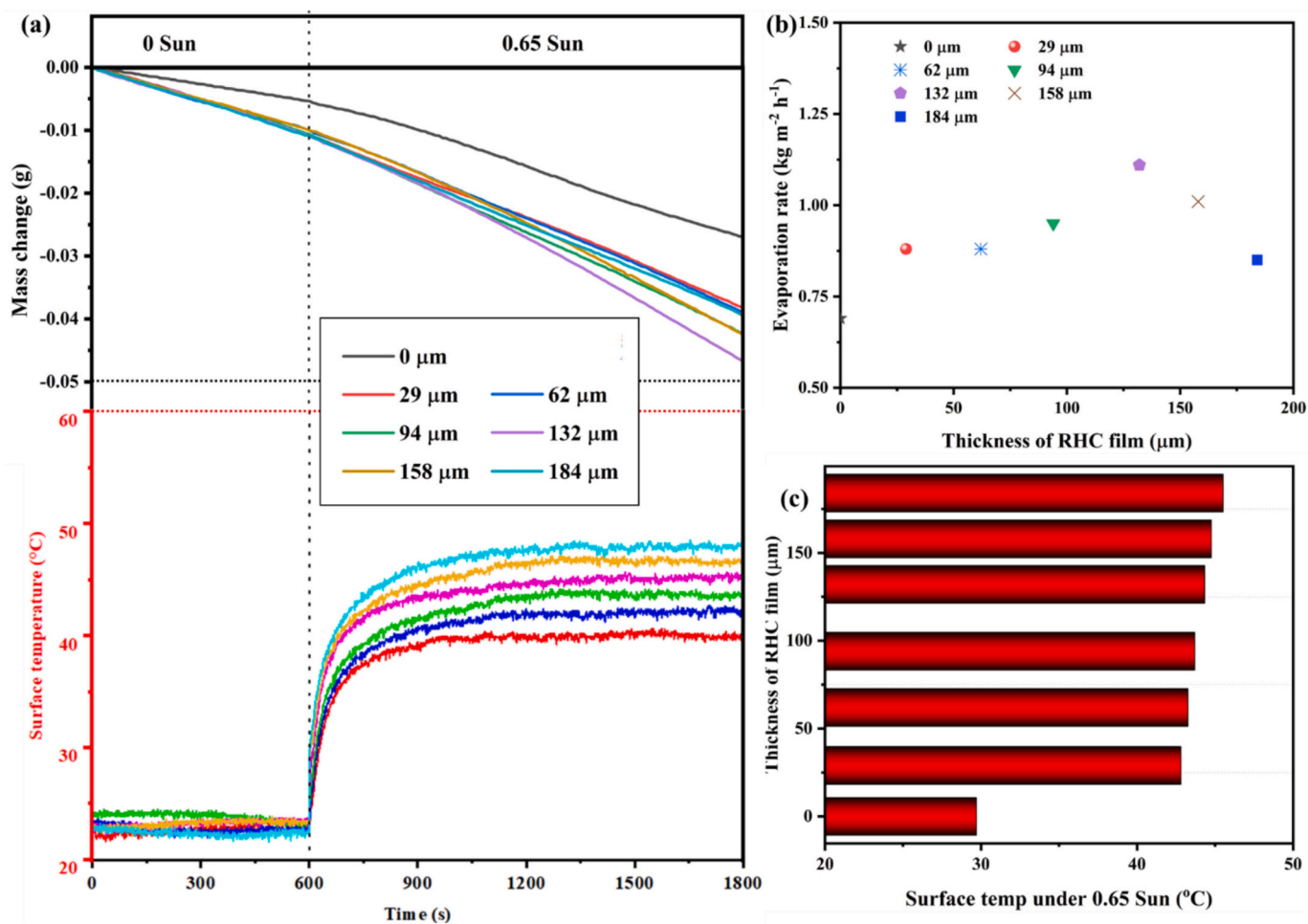


Fig. 6. (a) Mass change and surface temperature change over time, (b) Evaporation rate, and (c) Surface temperature of carbon-based floating system at 20 min, with the RHC thickness in the range of 0–184 μm .

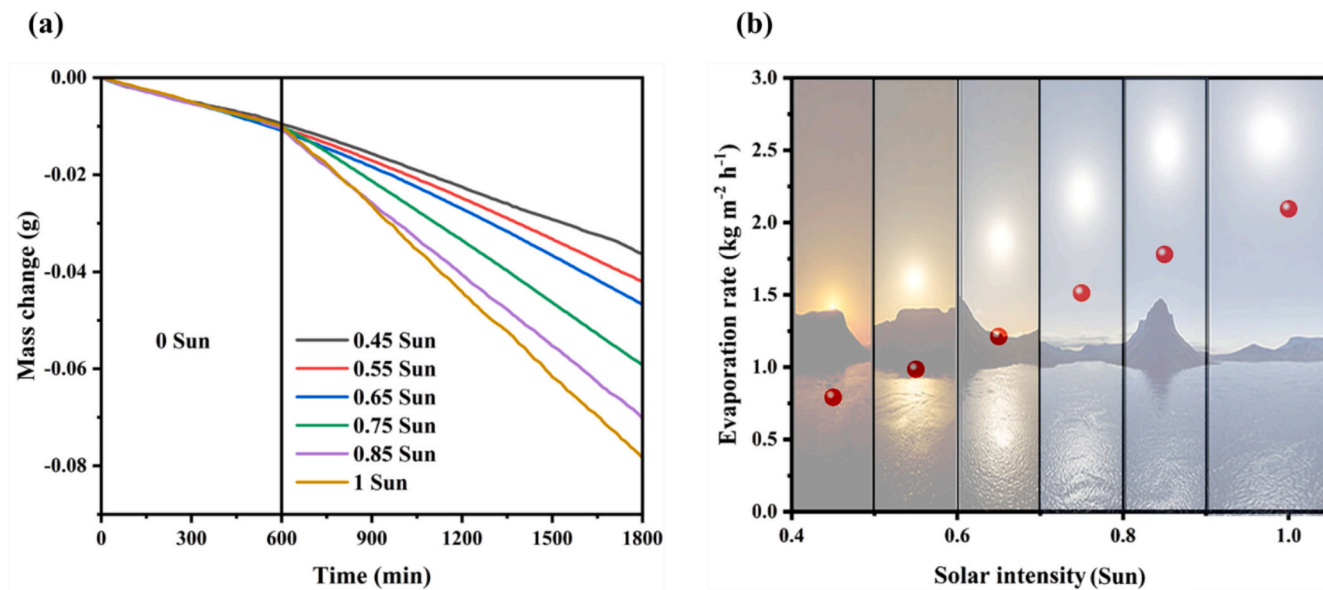


Fig. 7. (a) Mass change, and (b) Evaporation rate of 0.08 g carbon-based floating system under 0.45–1 sun.

Table 2

Evaporation rates of solar-driven steam-generation systems with different materials under 1 sun.

Photothermal materials	The structure of devices	Evaporation rate ($\text{kg m}^{-2} \text{h}^{-1}$)	Refs
Activated carbon	AC/Insulation foam	1.22	[40]
SnSe-SnSe ₂	SnSe-SnSe ₂ /Glassy carbon foam	1.28	[41]
Carbon nanotubes (CNT)	CNTs/Cellulose hydrogel	1.52	[42]
MWNTs	MWNTs/Cellulose/PS foam	1.48	[39]
MoS ₂ Nanosheet	MoS ₂ /carbon foam	1.49	[43]
Porous carbon nanofoam	Porous carbon nanofoam	1.68	[44]
Geopolymer -mesoporous carbon	Geopolymer - M. carbon/PS foam	1.58	[45]
Carbon dots	CDs/wood	2.27	[46]
Carbon nanotubes /polyvinyl alcohol hybrid gels	CNTs/PVA Gels	2.06	[47]
Carbonized coconut husk	3D configurations-based on carbonized coconut husk	3.6	[48]
MnO ₂ /Carbon black/Commercial acrylic paint	3D configurations-based on MnO ₂ /Carbon black/Commercial acrylic paint	1.4	[49]
Selective solar absorber -coated aluminum sheet	Cuboid-shaped ISSG-based on Selective solar absorber -coated aluminum sheet	1.9 (3.5 wt% of saltwater)	[50]
RHC	RHC/cellulose/PS foam	2.09	This study

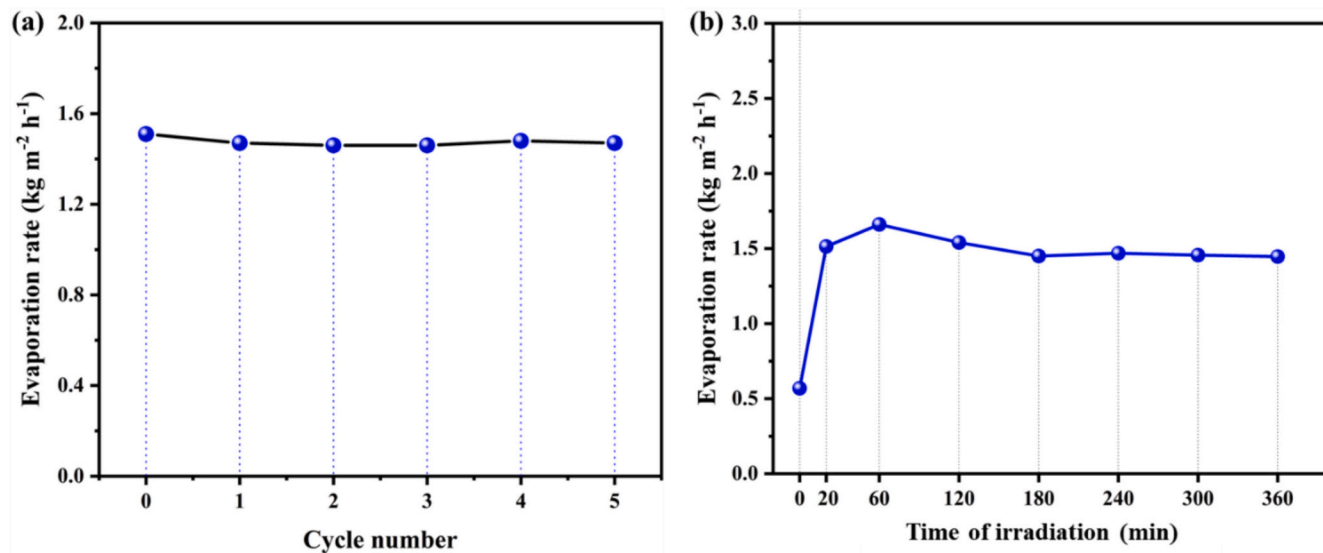


Fig. 8. (a) Reusability, and (b) durability of 0.08 g c-based floating system under 0.75 sun.

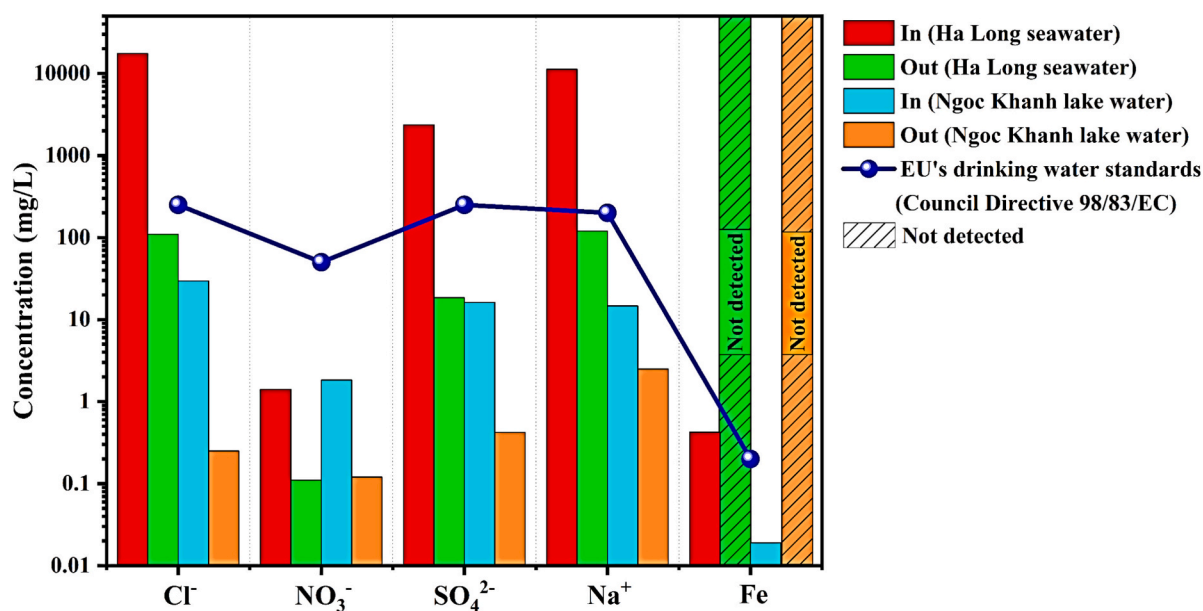


Fig. 9. Ion concentrations of two natural water samples before and after distillation.

To assess the feasibility of using an RHC-based membrane for pollutant removal and freshwater harvesting, we collected water samples from the condensed vapor generated by solar-driven evaporation. Water from Ngoc Khanh Lake and Ha Long Bay was used as the input. The concentrations of several major anions and cations, such as chloride, nitrate, sulfate, sodium, iron, and calcium, were analyzed.

Table 3
Water-quality analysis results before and after distillation.

Parameter	Sample			
	Ha Long Bay seawater		Ngoc Khanh Lake water	
	Input (mg/L)	Output (mg/L)	Input (mg/L)	Output (mg/L)
pH	8.4	6.6	7.6	6.4
Cl ⁻	17,470	109.62	29.6	0.25
NO ₃ ⁻	1.40	0.11	1.83	0.12
SO ₄ ²⁻	2357.50	18.52	16.14	0.42
Br ⁻	ND	ND	ND	ND
As (total)	ND	ND	ND	ND
Cd ²⁺	ND	ND	ND	ND
Pb ²⁺	ND	ND	ND	ND
Cr (total)	ND	ND	ND	ND
Cu ²⁺	ND	ND	ND	ND
Sb ²⁺	0.008	ND	0.002	ND
Zn ²⁺	0.193	ND	ND	0
Mn ²⁺	0.006	ND	0.014	ND
Al ³⁺	ND	ND	ND	ND
Ni ²⁺	0.001	ND	ND	ND
Fe (total)	0.423	ND	0.019	ND
Hg ²⁺	ND	ND	ND	ND
Ca ²⁺	452.50	3.78	15.148	1.537
Na ⁺	11,225.30	119.68	14.687	2.487

ND: Not detected.

As presented in Fig. 9 and Table 2, the concentrations of all analytical parameters in the distillate were significantly reduced to levels below the maximum allowed values set by Council Directive 98/83/EC on the quality of water intended for human consumption. Over 90 % of anions and cations present in the input samples were removed. Remarkably, the removal efficiency of chloride reached up to 99 % in both tests. Additionally, heavy metals at trace concentrations—namely Sb, Zn, Mn, Ni, and Fe—were below the limit of detection of the instrument.

The mechanisms of RHC-based treatment of pollutants and water

purification can be explained by the following ideas: (i) cellulose, with a pore size of 2.5 μm as a substrate, can efficiently remove particles and bacterial fragments [51]; (ii) anions and, especially, heavy metals are adsorbed by the RHC layer; (iii) salts and contaminants are retained on the surface of the absorber layer during the phase transition from liquid to vapor. The carbon-based solar distillation process has also been proven to be highly effective at removing bacteria and organic dyes [52].

4. Conclusion

In conclusion, this work highlights the remarkable capability of RHC as a high-performance material for solar steam generation, with further applications in freshwater production. By developing a solar absorber layer based on RHC, a carbon material derived from agricultural waste, we assessed its photothermal conversion performance. The RHC exhibited strong solar absorption, exceeding 80 % in the 200–2250 nm range and demonstrated excellent hydrophilicity, which facilitates effective water transport—both essential characteristics of photothermal materials. The RHC-based ISSG system achieved a maximum steam generation rate of 2.09 $\text{kg m}^{-2} \text{h}^{-1}$ under standard solar irradiation (1 sun), along with outstanding operational stability. Furthermore, the system successfully removed approximately 85 % of ions from the water post-evaporation. These promising outcomes suggest that RHC, being both renewable and cost-effective, holds significant potential for seawater desalination and wastewater treatment. Despite challenges in quantifying the efficiency of each process, the overall evidence confirms RHC's strong potential in clean-water production, solar desalination, and water purification technologies.

CRedit authorship contribution statement

Thi Hien Nguyen: Methodology, Investigation, Data curation, Formal analysis, Writing – Original Draft. **Nam Anh Tran:** Methodology, Investigation, Data curation. **Thi Bich Ngoc Nguyen:** Data collection, Investigation, Experimental support. **Thi My Duyen Bui:** Data curation, Formal analysis. **Phuong Anh Lam:** Methodology, Formal analysis. **Ho-Suk Choi:** Data curation, Formal analysis. **Dieu Linh Do:** Instrumentation, Characterization, Validation. **Thi Hanh Nguyen:** Data curation, Formal analysis. **Viet Phuong Hoang:** Data curation, Formal analysis. **Hai Linh Nguyen:** Data curation, Formal

analysis. **Thuy Duong Pham:** Methodology, Formal analysis. **Van-Duong Dao:** Conceptualization, Supervision, Project administration, Writing – Review & Editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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